

Journal Pre-proof

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PII: S2214-8450(25)00177-2

DOI: <https://doi.org/10.1016/j.bir.2025.10.026>

Reference: BIR 753

To appear in: *Borsa Istanbul Review*

Received Date: 18 July 2025

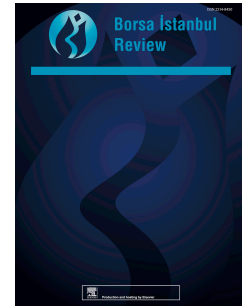
Revised Date: 25 September 2025

Accepted Date: 24 October 2025

Please cite this article as: Wang G., Biodiversity risk and firms' green investment: Evidence from China, *Borsa Istanbul Review*, <https://doi.org/10.1016/j.bir.2025.10.026>.

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Biodiversity risk and firms' green investment: Evidence from China

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ABSTRACT

Amid growing global concerns over biodiversity risk and its financial implications, understanding how firms strategically respond to biodiversity risk has become a critical issue in sustainable finance. This study seeks to answer the research question: Does rising biodiversity risk stimulate or deter firms' green investment? This is one of the first papers to examine firms' green investment decisions during biodiversity risk exposure, using Chinese listed firms' data from 2011 to 2023. Employing a fixed-effect model, we find that firms exposed to biodiversity risk increase their green investment by approximately 60.640%, compared to firms not exposed to such risk. Furthermore, moderating effect analysis indicates that managerial shareholdings and corporate ESG performance can enhance the positive effect of biodiversity risk on firms' green investment. Additionally, heterogeneity analysis reveals that this positive effect is more pronounced in SOEs and pollution-intensive industries. Overall, we provide empirical support for the strategic growth option theory, suggesting that firms follow a preemptive strategy during ecological risk exposure.

Keywords:

Biodiversity risk

Firms' green investment

Managerial shareholdings

Corporate ESG performance

Moderating effect

1. Introduction

Biodiversity risk poses a significant threat to global economic stability, with potential annual losses estimated at \$20 trillion (Giglio et al., 2023). This issue has

rapidly risen to prominence on both the corporate and regulatory agendas in China. Chinese listed firms are increasingly exposed to biodiversity-related risks (He et al., 2024), and corporate managers are becoming more aware of their potential impacts on corporate investment (Pi et al., 2025). Regulatory responses have also intensified. Notably, the 2024 issuance of the “Self-Regulatory Guidelines for Listed Companies No. 14—Sustainable Development Report (Trial)” mandates biodiversity reporting for the first time, reflecting and reinforcing a growing managerial awareness of the issue. Therefore, it is crucial to understand how Chinese businesses cope with biodiversity risk.

Firms face a dual threat from biodiversity loss. Beyond the immediate physical risks like habitat destruction, companies must now navigate a complex web of transition risks. These include new regulatory landscapes, evolving market dynamics, and reputational pressures, all of which are fundamentally altering corporate valuations and strategic planning (Pi et al., 2025). Firstly, tightening environmental regulations and increasingly stringent carbon emission constraints are raising both operational costs and compliance burdens for firms (Li et al., 2025). Secondly, rising societal and consumer expectations are intensifying scrutiny of firms’ ecological footprints (Kopnina et al., 2024), posing significant risks to brand reputation and long-term competitive advantage. Although some recent studies suggest that biodiversity risk may improve corporate ESG performance (Pi et al., 2025) or promote digital transformation (Wang, 2025), the majority of existing research indicates that biodiversity risk predominantly poses significant threats to firms’ operational activities (Ahmad & Karpuz, 2024; Liang et al., 2024; Moses & Yahaya, 2024; Yang & Li, 2025).

Risks and uncertainties inevitably influence corporate investment behavior. Effective risk management is therefore important for firms to navigate environmental uncertainty and help mitigate potential financial losses (Li et al., 2025). In this context, green investment plays a crucial role in addressing biodiversity risk (Balsalobre Lorente et al., 2023). Green investment, such as developing new technologies for industrial waste recycling or creating more efficient clean energy systems, serves as strategic tools for mitigating ecological impacts. However, green investment shares key characteristics

with traditional research and development (R&D) (Chen & Ma, 2021), such as high uncertainty, long time horizons, and significant irreversibility (Siedschlag & Yan, 2021), making it particularly sensitive to external shocks. **Relying on real options and strategic growth option theories**, scholars offer two opposing views on whether risk deters or stimulates corporate investment. Prior research has explored the impact of external risk factors—such as climate risk (Wang & Wang, 2025), political risk (Boah & Ujah, 2024), market risk (Lai et al., 2015), and legal risk (Xiao, 2013)—on traditional R&D investment, providing a rich theoretical basis for understanding investment decisions under uncertainty. However, there remains a gap in understanding how biodiversity risk specifically shapes corporate green investment strategies. Amid growing global concerns over biodiversity risk and its financial implications, understanding how firms strategically respond to such ecological pressure has become a critical issue in sustainable finance. This study seeks to answer the core research question: Does rising biodiversity risk stimulate or deter firms' green investment?

This paper utilizes panel data from Chinese listed firms between 2011 and 2023 to explore the relationship between biodiversity risk and firms' green investment. Our regression analysis provides robust evidence that biodiversity risk promotes firms' green investment. Furthermore, moderating effect analysis indicates that managerial shareholdings and corporate ESG performance can enhance the positive effect of biodiversity risk on firms' green investment.

This study makes several novel contributions:

First, this is one of the first papers to examine firms' green investment decisions during biodiversity risk exposure. While many of the existing studies view environmental risks as operational burdens or financial threats to corporate activities (Adamolekun, 2024; Ahmad & Karpuz, 2024; Yang & Li, 2025), our findings reveal a more nuanced reality: firms strategically increase green investment when faced with this risk. This provides compelling evidence of a proactive, forward-looking corporate response, moving beyond a purely defensive or threat-based perspective.

Second, we resolve a key theoretical tension regarding investment under ecological uncertainty. By empirically testing two competing real options

frameworks—one predicting investment delay under uncertainty, and the other anticipating preemptive investment to capture first-mover advantages—this study provides clear and robust support for the strategic growth option theory in the context of biodiversity risk. We demonstrate that for ecological pressures, the strategic value of preemptive investment outweighs the value of waiting, thereby extending the applicability of strategic growth option theory to the vital domain of corporate sustainability.

Third, we identify internal governance mechanisms as critical enablers of strategic responsiveness. Moderating analysis shows that managerial shareholdings and strong ESG performance do not merely enable but actively amplify a firm's capacity to convert ecological risks into strategic green investments. We bridge the literature on corporate governance and sustainability, by demonstrating how firms translate external risks into strategic action. This highlights a critical pathway through which internal governance fosters corporate resilience and proactivity.

Finally, our findings offer a granular and actionable roadmap for targeted policymaking, particularly in emerging economies like China. Heterogeneity analysis reveals that the positive response to biodiversity risk is significantly stronger among state-owned enterprises (SOEs) and pollution-intensive firms. This suggests that policies like green financing incentives, subsidies for eco-innovation or mandatory biodiversity disclosure requirements can be most effectively focused on these specific segments to accelerate a nationwide green transition, maximizing policy impact.

2. Literature review and hypothesis development

2.1. Biodiversity risk and firms' green investment

Firms' green investment is inherently risky, characterized by continuous capital outlays and irreversibility, and typically requires substantial and sustained financial support (Siedschlag & Yan, 2021). The presence of biodiversity risk introduces additional layers of uncertainty that can significantly influence firms' strategic investment decisions (Li et al., 2025).

Real options theory suggests that as uncertainty increases, the value of the option

to wait becomes greater, thus discouraging corporate investment in the near term (Myers, 1977). Green investment, in particular, shares many characteristics with traditional research and development (R&D) (Chen & Ma, 2021), such as high uncertainty, long-term horizons, and significant irreversibility (Siedschlag & Yan, 2021). Under conditions of uncertainty, firms tend to place greater value on the option to wait, rather than committing to irreversible and costly investments in the short term (Vo & Le, 2017). Consequently, the higher the degree of investment irreversibility, the stronger the incentive for firms to defer investment. In addition, the risk of project failure or low returns increases during periods of uncertainty. Given the inherent uncertainty and sunk-cost nature of many green projects (Liu et al., 2022), firms may choose to delay their investment decisions when uncertainty is high (Chronopoulos et al., 2016), significantly increasing the value of the postponement option. As a result, under the pressure of uncertainty, firms may prioritize the option to delay over the option to grow.

Empirical studies support this view, indicating that biodiversity risk may threaten firm performance by reducing operational efficiency (Li et al., 2025), intensifying maturity mismatch in financing and investment (Yang & Li, 2025), and increasing stock price crash risk (Liang et al., 2024). These factors that collectively threaten operational stability (Giglio et al., 2023). Moreover, biodiversity risk increases operating costs and reduces profitability (White et al., 2023), thereby weakening firms' internal financial capacity for sustained strategic investments, including green projects. In addition, a substantial body of research confirms a negative relationship between uncertainty and investment activity (Arif Khan et al., 2020; Boah & Ujah, 2024; Choi et al., 2022; Vo & Le, 2017).

Thus, drawing on the perspective from real options theory regarding the value of deferral, we propose the following hypothesis:

Hypothesis 1a. Biodiversity risk negatively affects firms' green investment.

In contrast, **the strategic growth option theory** posits that firms are more inclined to invest aggressively during periods of uncertainty. This perspective, often

summarized by the proverb, “The early bird catches the worm,” assumes the presence of imperfect competition where first-mover advantages are significant (Kulatilaka & Perotti, 1998). In such environments, postponing investment may enable competitors to seize valuable opportunities, whereas immediate action can deter entrants and enhance a firm’s competitive advantage—making proactive investment a strategic necessity.¹

Empirical studies support this viewpoint. Stein & Stone (2013) observed a positive relationship between uncertainty and innovation-related investment. However, they also note that the global financial crisis between 2008 and 2010 led to a decline in capital investment worldwide, attributing this drop to heightened economic uncertainty. Vo & Le (2017) find that firms in highly competitive industries with weaker market power are more likely to increase R&D investment during periods of uncertainty. Recent studies further extend this line of inquiry to environmental contexts. Tian et al. (2024) show that climate risk positively influences firms’ engagement in green innovation. Similarly, Dutta et al. (2023) demonstrate that increasing climate risk encourages investment in alternative energy sources. This growing demand for green energy not only raises the value of related investments but also reduces their volatility, thus reinforcing the relevance of strategic growth option theory in the context of sustainability-driven decision-making under uncertainty.

This theoretical lens is particularly applicable in today’s China, where the government is increasingly emphasizing “ecological civilization” and sustainable development, biodiversity risk is not just a threat but also a key indicator in current environmental regulation. Therefore, faced with the uncertainty posed by increasing biodiversity risk, firms have a stronger incentive to adopt a preemptive investment strategy. This context amplifies the value of strategic green investments, a rationale supported by several interconnected theories.

On one hand, from the perspective of **stakeholder theory**, firms face heightened

¹ As a notable example, the COVID-19 global pandemic prompted pharmaceutical companies to accelerate R&D expenditures aimed at understanding the virus and advancing vaccine development. And this proactive investment was reflected in their rising stock prices. For example, stock price of *Pfizer Inc.* rose from \$30 at the end of 2020 to \$50 at the end of 2022. As a global pharmaceutical company known for developing innovative medicines and vaccines, *Pfizer Inc.* exemplifies how timely investment in R&D can yield significant returns.

pressure to meet societal expectations to maintain their legitimacy and social acceptance (Hannan & Freeman, 1984). Specifically, biodiversity risk introduces significant complexity and unpredictability, which in turn compels firms to invest more in innovative technologies that enhance their agility, responsiveness, and adaptive capacity. Proactive green investment becomes a crucial tool for managing these stakeholder relationships, and align with national strategic priorities, thus turning risk into a competitive advantage. On the other hand, drawing on the **dynamic capabilities theory**, firms must continuously develop new competencies to navigate such environmental uncertainties. Importantly, while green investments may not always result in formal patents, they often lead to proprietary process improvements, unique market positioning, or other forms of competitive advantage that can be defended. As Bloom & Van Reenen (2002) argue that the ability to monetize such advantages (e.g., through the sale of intellectual property rights from R&D) mitigates the inherent irreversibility of the initial spending, thereby encouraging firms to invest even during periods of uncertainty.

Collectively, these theories suggest that when faced with biodiversity risk, the strategic value of preemptive action—securing market position, satisfying stakeholders, and building new capabilities—outweighs the value of waiting. Thus, we propose the competing hypothesis:

Hypothesis 1b. Biodiversity risk can promote firms' green investment.

2.2. Moderating effect of managerial shareholdings

According to agency theory, managers often exhibit risk-averse and short-term tendencies in investment decision-making due to their inability to fully diversify personal financial risk associated with their own wealth portfolios (Wright et al., 1996). This aversion to risk may lead to underinvestment in high-risk, high-reward initiatives such as strategic green projects (Addoum et al., 2020). Corporate governance mechanisms are therefore crucial for mitigating this agency conflict. Managerial shareholdings, as a key equity-based incentive mechanism, can alleviate such tendencies by aligning the financial interests of managers with those of shareholders

(Fu et al., 2024). When managerial incentives are closely linked to firm performance, managers become more inclined to pursue long-term strategic investments—including those in green investment—despite the associated risks (Wu et al., 2022).

This alignment encourages a longer-term perspective and is expected to positively moderate a firm's investment response to biodiversity risk through two primary and reinforcing channels:

First, from a long-term value creation perspective, which aligns with the strategic growth option theory, equity incentives encourage managers to transcend the immediate costs and uncertainties of green projects. Rather than delay investment, they are more motivated to identify and seize future growth options, viewing green investment as a critical strategy to build a sustainable competitive advantage and enhance the firms' long-term value. Second, from a risk mitigation and reputational management perspective, managers with significant shareholdings have their personal wealth directly tied to the firm's market performance. They are therefore highly sensitive to factors that could trigger reputational damage or stock price volatility, such as inadequate responses to biodiversity-related issues. In this scenario, green investment becomes not only a growth strategy but also an essential tool for managing risk and safeguarding the firm against negative stakeholder scrutiny.

By operating through both the pursuit of strategic opportunities and the mitigation of personal and corporate risk, managerial shareholdings can effectively transform external ecological pressure into a powerful internal driver for proactive green investment. Therefore, we propose the following hypothesis:

Hypothesis 2. Managerial shareholdings positively moderate the relationship between biodiversity risk and firms' green investment.

2.3. Moderating effect of corporate ESG performance

Firms with a strong track record of corporate ESG performance are better positioned to translate external pressures, such as biodiversity risk, into strategic green investments (Pi et al., 2025). This moderating effect operates through several complementary channels grounded in stakeholder, signaling, and resource-based

theories, enabling firms to implement proactive risk mitigation strategies.

First, according to **signaling theory**, high ESG performance serves as a credible signal to capital markets of the firms' superior management quality and stability. This enhanced transparency helps reduce information asymmetry, enabling high-ESG firms to gain greater investor trust and secure external financing with greater ease (Shang et al., 2025). This improved access to external capital reduces reliance on internal funds (Eccles et al., 2014), thereby providing the necessary financial slack for firms to undertake significant, long-term green investments.

Second, from the perspective of **stakeholder theory** and organizational legitimacy, operational disruptions stemming from biodiversity risk intensify stakeholder pressure and may even jeopardize a firm's social legitimacy (Lähtinen et al., 2016). As external expectations for corporate environmental responsibility increase, firms are incentivized to enhance their ESG performance to maintain legitimacy and gain stakeholder support (Hannan & Freeman, 1984). This reservoir of goodwill acts as a buffer against reputational threats and facilitates smoother execution of green projects.

Finally, through the lens of the **Resource-Based View (RBV)**, a high ESG rating is not merely a public relations asset but a manifestation of a firm's underlying dynamic capabilities. It reflects the development of sophisticated internal processes for identifying, managing, and integrating complex non-financial risks into its core strategic decision-making. Consequently, when confronted with an emerging challenge such as biodiversity risk, these firms are inherently more competent and agile in formulating and executing a strategic response, such as a targeted green investment plan.

Collectively, these channels create a virtuous cycle: ESG performance enhances financial capacity, builds stakeholder support, and signals superior internal capabilities. This combination empowers firms to more effectively convert the challenge of biodiversity risk into a strategic opportunity for green investment. Therefore, we propose the following hypothesis:

Hypothesis 3. Corporate ESG performance positively moderates the relationship between biodiversity risk and firms' green investment.

The theoretical framework of our empirical study is represented in **Fig. 1**.

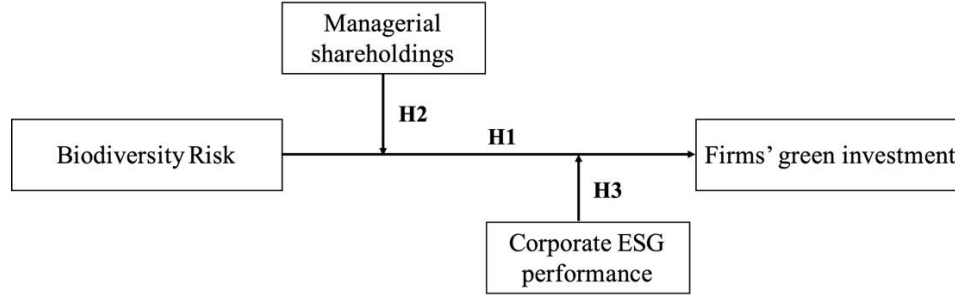


Fig. 1. Theoretical framework.

3. Research design

3.1. Methodology

To test the primary hypothesis (H1) that biodiversity risk influences firms' green investment, we employ a fixed-effect panel regression model, specified as follows in Eq. (1):

$$GI_{it} = \alpha + \beta BR_{it-1} + \sum \gamma \cdot Controls_{it} + year_t + firm_i + \epsilon_{it} \quad (1)$$

To examine the proposed moderating effects (H2 and H3), we augment the baseline model with an interaction term, as specified in Eq. (2):

$$GI_{it} = \alpha + \beta BR_{it-1} + \delta Moderators_{it-1} + \theta BR_{it-1} * Moderators_{it-1} + \sum \gamma \cdot Controls_{it} + year_t + firm_i + \epsilon_{it} \quad (2)$$

Where the Moderators represent our two moderating variables: managerial shareholdings (MS) and corporate ESG performance (ESG).

3.2. Variable Definition

3.2.1. Dependent variable: firms' green investment

Following Chen & Ma (2021) and Eyraud et al. (2013), we use firms' green investment expenditures (GI) as dependent variable. GI is measured as the natural logarithm of a firm's total capital expenditure related to environmental protection, plus one. GI is manually sorted and collected from firms' annual reports disclosed by Cninfo.²

These expenditures cover a broad range of environmental protection activities, representing a firm's overall capital commitment to a green transition, which often entails the adoption and implementation of new technologies and processes. Therefore,

² Website: <http://www.cninfo.com.cn/new/index>.

they serve as a suitable proxy for green investment. The types of firms' green investment expenditures are shown in **Fig.2**.

	Types of green investment expenditures	Detail description
1	Energy Conservation, Emission Reduction, and Energy Structure Optimization	Energy conservation, Emission reduction, Oil-to-gas conversion, Coal-to-gas conversion, Ozone technology upgrade, New energy, Hydrogen
2	Industrial Waste and Environmental Pollution Control	Wastewater, Waste gas / Exhaust gas, Slag / Solid waste residue, Dust removal / Dust control, Pollution / Contamination, Garbage / Waste, Waste materials, Tailings, Acid mist, Phosgene, Dust emissions / Particulate matter, Drainage, Backfilling, Exhaust gas treatment, Disposal / Abandonment, Recovery, Recycling / Resource recovery, Circular utilization / Resource reuse
3	Air Pollution Control and Combustion Equipment Upgrading	Desulfurization, Denitrification / NO _x removal, Boiler retrofitting / Boiler upgrading
4	Clean Energy and Resource Recycling System Construction	Wind power, Photovoltaics / Solar PV, Clean energy, Green (in context of environment), Circular economy, Cogeneration / Thermal power, Natural gas, River and Pond Energy Convergence Power Station
5	Mine Ecological Restoration and Environmental Comprehensive Management	Ecological restoration, Environmental safety, Comprehensive management / Integrated remediation, Water pollution prevention and control, Water source improvement / Water supply system upgrading, Restoration / Remediation
6	Urban and Ecological Greening Construction	Afforestation / Greening, Vegetation construction, Tree planting / Reforestation, Green space maintenance

Fig. 2. The types of firms' green investment expenditures

In addition, to objectively visualize the trend in green investment intensity, independent of firm size or inflation, we construct a “green investment ratio”³ to illustrate this trend. As is shown in **Fig.3**, the average green investment ratio has exhibited a fluctuating upward trend from 2011 to 2023. This descriptive evidence suggests that green investment has become a more salient component of corporate capital allocation.

³ The green investment ratio is calculated as the ratio of green investment amount to total assets, multiplied by 100.

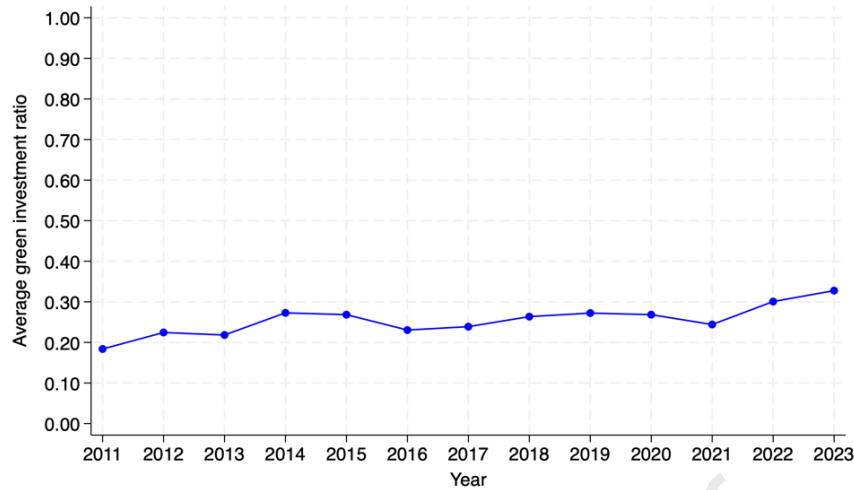


Fig. 3. Time plot for firms' average green investment ratio

3.2.2. Independent variable: Biodiversity risk

The independent variable, **biodiversity risk (BR)**, is measured using the firm-level index for Chinese listed companies developed by He et al. (2024). **The dataset is publicly accessible,⁴** and has been widely adopted in recent studies (eg., Li et al., 2025; Pi et al., 2025; Wang, 2025).

We selected this index as it is the most comprehensive and current publicly available measure of corporate biodiversity exposure in China. Its construction is methodologically sound, based on a textual analysis of corporate annual reports using a widely recognized biodiversity dictionary from Giglio et al. (2023).

Following the original study, BR equals 1 if the frequency of biodiversity-related terms in a firm's annual report is more than two, and 0 otherwise. These biodiversity-related terms are listed in detail in Appendix Table A.1. As is shown in **Fig.4**, the average level of firm-level biodiversity risk has exhibited a continuously upward trend from 2011 to 2023. This sustained increase aligns closely with the trajectory reported in He et al. (2024).

⁴ Website: <https://sites.google.com/view/brianmlucey/research?authuser=0>

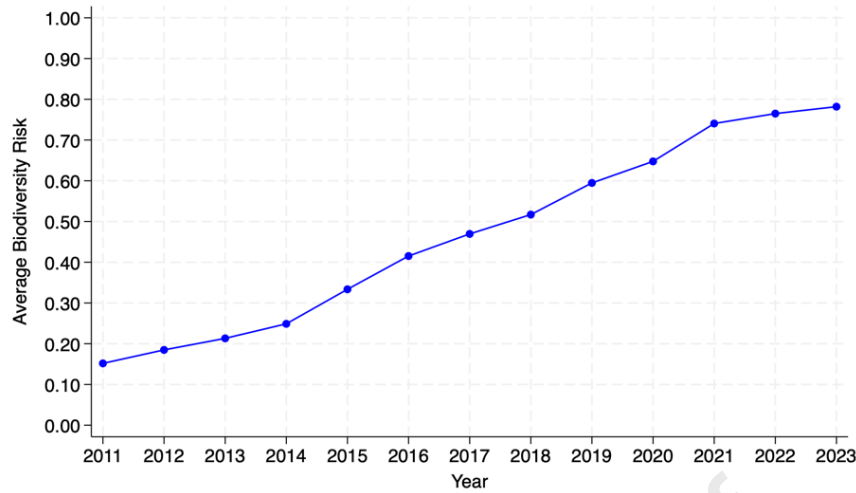


Fig. 4. Time plot for firms' average biodiversity risk

3.2.3. Moderating variable: Managerial shareholdings and corporate ESG performance

We use Sino-Securities Index (SSI) ESG ratings as a proxy for corporate ESG performance. Based on SSI database, corporate ESG ratings are assigned from 1(C rating) to 9(AAA rating). Managerial shareholdings is calculated as the natural logarithm of total number of share held by the firm's directors, senior management, and supervisors, plus one.

3.2.4. Control variables

We select 10 control variables that may influence firms' green investment. To control for other factors that could influence firms' green investment, we select the following control variables. Firm size (SIZE), leverage (LEV), and return on assets (ROA) are included as basic financial characteristics that affect a firm's investment capacity and capital structure. Firm growth (GROWTH) and Tobin's Q (TOBINSQ) reflect a company's future growth potential and market value, which may drive its investment decisions. Corporate governance variables, such as ownership concentration (TOP1) and CEO-Chair duality (DUAL), can influence managerial decision-making efficiency and risk appetite. Firm age (AGE) and state ownership (SOE) represent the firm's experience and the social responsibilities it may bear, respectively. Finally, the gross domestic product per capita (GDP) of the firm's province is included to control for the influence of the macroeconomic environment on corporate investment behavior.

Detailed definition of variables is presented in appendix Table A.2.

3.3. Data sources and sample selection process

Our research sample comprises Chinese A-share listed companies on the Shanghai and Shenzhen stock exchanges. We constructed our dataset from multiple sources: Data on firms' green investment (GI) were manually sorted and collected from firms' annual reports disclosed by Cninfo.⁵ Firms' financial data were drawn from the CSMAR database. Biodiversity risk (BR) metrics were adopted from the public index developed by He et al. (2024). ESG ratings were gathered from Sino-Securities Index ESG ratings database. Samples are preprocessed in following criteria: (1) excluding ST, *ST, PT firms, and firms in the delisting consolidation period; (2) excluding firms in financial industry; (3) excluding firms with missing data; (4) applying a 1% winsorization to numerical variables to mitigate the impact of outliers. Detailed definitions for all variables are provided in Appendix Table A.2.

Ultimately, 25,998 firm-year observations from 2011 to 2023 are obtained. In the regression analysis, the final number of effective observations is 23,203 due to the use of lagged variables and the requirement of complete data for all variables in the model.

⁵ Website: <http://www.cninfo.com.cn/new/index>.

4. Empirical analysis

4.1 Summary statistics

4.1.1 Variable descriptive statistics

Table 1

Descriptive statistics of variables

	Obs	Mean	Std.	Min	Max
GI	25,998	5.442	7.769	0.000	21.070
BR	25,998	0.498	0.500	0.000	1.000
SIZE	25,998	1.842	4.711	0.062	34.300
LEV	25,998	0.432	0.193	0.064	0.852
ROA	25,998	0.043	0.049	-0.132	0.196
GROWTH	25,998	0.055	0.081	-0.276	0.321
DUAL	25,998	0.252	0.434	0.000	1.000
TOBINSQ	25,998	1.850	1.080	0.820	7.059
AGE	25,998	2.946	0.343	1.792	3.584
TOP1	25,998	0.361	0.153	0.088	0.761
SOE	25,998	0.414	0.493	0.000	1.000
GDP	25,998	11.230	0.495	9.995	12.160

4.1.2 Correlation analysis

Table 2 presents the results of the Pearson correlation analysis for the main variables in this study.

Table 2

Result of Pearson correlation analysis

	GI	BR	SIZE	LEV	ROA	GROWTH	DUAL	TOBINSQ	AGE	TOP1	SOE	GDP
GI	1.000											
BR	0.106***	1.000										
SIZE	0.082***	0.169***	1.000									
LEV	0.116***	0.092***	0.313***	1.000								
ROA	-0.019***	-0.060***	-0.048***	-0.391***	1.000							
GROWTH	0.022***	-0.032***	0.066***	-0.056***	0.804***	1.000						
DUAL	-0.079***	0.006	-0.080***	-0.139***	0.067***	0.005	1.000					
TOBINSQ	-0.105***	-0.089***	-0.192***	-0.307***	0.265***	0.138***	0.078***	1.000				
AGE	0.044***	0.228***	0.040***	0.141***	-0.138***	-0.087***	-0.064***	-0.056***	1.000			
TOP1	0.045***	-0.002	0.155***	0.020***	0.125***	0.092***	-0.052***	-0.058***	-0.152***	1.000		
SOE	0.121***	0.036***	0.227***	0.270***	-0.119***	0.006	-0.314***	-0.129***	0.129***	0.207***	1.000	
GDP	-0.081***	0.253***	0.139***	-0.029***	-0.038***	-0.064***	0.116***	-0.036***	0.286***	0.004	-0.128***	1.000

4.2. Baseline regression

Column (1) and (2) of Table 3 present baseline regression results. The coefficients of BR are significantly positive at the 1 % level, regardless of whether control variables are included.

In column (2), the coefficient of BR (0.474) is not only statistically significant but also economically meaningful. As the dependent variable is in natural logarithm form, this coefficient implies that, ceteris paribus, firms exposed to biodiversity risk (BR=1) increase their green investment by approximately 60.640% ($e^{0.474}-1$)⁶ compared to firms not exposed to such risk. This indicates a substantial investment response to perceived biodiversity challenges.

This finding supports the strategic growth option theory (Hypothesis 1b), suggesting that firms view biodiversity risk as a catalyst for strategic investment rather than a reason to defer it.

⁶ Since the dependent variable (GI) is in natural logarithm form and the independent variable (BR) is a dummy variable, the precise percentage change in the dependent variable is calculated as $(e^{\beta}-1) \times 100\%$. In this case, with $\beta=0.474$, the effect is $(e^{0.474}-1) \approx 0.6064$, or 60.640%.

Table 3

Results of baseline regression.

	(1)	(2)
	GI	GI
BR	0.505*** (3.966)	0.474*** (3.727)
SIZE		0.051 (1.038)
LEV		0.922 (1.387)
ROA		-3.430 (-1.350)
GROWTH		3.610*** (2.733)
DUAL		0.317* (1.850)
TOBINSQ		-0.072 (-1.187)
AGE		0.061 (0.053)
TOP1		1.462 (1.270)
SOE		0.408 (0.946)
GDP		0.217 (0.214)
Constant	4.278*** (24.150)	0.544 (0.048)
Year FE	YES	YES
Firm FE	YES	YES
Observations	23,203	23,203
Adjusted R ²	0.014	0.016

Note: ***, **, and * denote significance at 1 %, 5 %, and 10 % levels.

Numbers in parentheses are robust t-value.

4.3. Robustness test

4.3.1. Changing the measurement approach of dependent variable

We use GI_1⁷ and GI_2⁸ as an alternative measurement of GI. The result in column (1) and column (2) of Table 4 is consistent with that of Table 3, confirming that the baseline regression is robust.

Table 4

Alternative measurement of GI

	(1) GI_1	(2) GI_2
BR	0.070*** (4.030)	0.192*** (4.359)
SIZE	0.005 (1.021)	0.014 (1.326)
LEV	0.199** (1.972)	0.524* (1.913)
ROA	-0.375 (-1.080)	-1.549* (-1.849)
GROWTH	0.343** (2.016)	0.581 (1.359)
DUAL	0.029 (1.128)	0.054 (0.872)
TOBINSQ	0.003 (0.351)	-0.007 (-0.342)
AGE	-0.248 (-1.437)	-0.461 (-1.014)
TOP1	0.172 (1.088)	0.530 (1.308)
SOE	0.079 (1.112)	0.268 (1.594)
GDP	-0.006 (-0.051)	-0.060 (-0.177)
Constant	0.694 (0.516)	1.734 (0.463)
Year FE	YES	YES
Firm FE	YES	YES
Observations	23,203	23,203
Adjusted R ²	0.003	0.004

⁷ GI_1 is defined as green investment as a percentage of total assets, calculated as (Green Investment expenditures/ Total Assets) × 100.

⁸ GI_2 is defined as green investment as a percentage of total assets, calculated as (Green Investment expenditures/ Operating income) × 100.

4.3.2. Sensitivity test of independent variable

We further assess the sensitivity of the independent variable (BR) to its threshold. Specifically, we use two variables: BR_one (set to 1 if the frequency of biodiversity-related terms in a firm's annual report is > 1 , and 0 otherwise) and BR_three (if the frequency of related terms is > 3 , and 0 otherwise), to run regression again. The test results are shown in columns (1) of Table 5, the coefficient for BR_one remains positive and significant. In columns (2), the coefficient for BR_three is also positive but not statistically significant at conventional levels (t -value = 1.370). These results confirm the robustness of our main findings.

Table 5

Alternative measurement of BR

	(1)	(2)
	GI	GI
BR	0.444*** (3.642)	0.187 (1.370)
SIZE	0.053 (1.062)	0.053 (1.075)
LEV	0.934 (1.407)	0.943 (1.421)
ROA	-3.342 (-1.312)	-3.485 (-1.366)
GROWTH	3.573*** (2.695)	3.648*** (2.755)
DUAL	0.319* (1.850)	0.321* (1.869)
TOBINSQ	-0.068 (-1.135)	-0.069 (-1.152)
AGE	-0.012 (-0.010)	-0.033 (-0.029)
TOP1	1.481 (1.284)	1.528 (1.324)
SOE	0.423 (0.976)	0.426 (0.986)
GDP	0.279 (0.274)	0.329 (0.322)
Constant	-0.045 (0.004)	-0.409 (-0.036)
Year FE	YES	YES
Firm FE	YES	YES
Observations	23,143	23,143
Adjusted R ²	0.016	0.015

4.3.3. 2SLS regression

We adopt *one-period lagged average biodiversity risk in the same industry* (IV-BR) as an instrumental variable for a firm's own biodiversity risk (BR), to run two-stage least squares (2SLS) regression.

A valid instrument must satisfy both the relevance and exogeneity conditions. The

exclusion restriction requires that the instrument affects a firm's green investment only through its impact on the firm's biodiversity risk. This is plausible in our setting: the lagged industry-average risk disclosure likely influences firms' current disclosure decisions due to peer effects or industry-wide scrutiny, but it is unlikely to directly determine a specific firm's present-day investment allocation.

The first-stage regression, reported in Column (1) of Table 6, confirms the instrument's relevance. Specifically, the coefficient on our instrument (IV-BR) is positive and highly significant, and the Cragg–Donald Wald F-statistic of 1467.637 far exceeds the conventional critical value of 10, mitigating concerns about a weak instrument problem. The second-stage estimation, shown in Column (2), reveals that the coefficient for BR is 1.918 and remains significant at the 1% level. This outcome suggests that our baseline finding is robust

Table 6

Results of 2SLS regression.

	(1)	(2)
	First stage	Second stage
	BR	GI
BR		1.918*** (4.909)
IV-BR	0.315*** (30.010)	
SIZE	0.006*** (4.582)	0.042 (1.555)
LEV	-0.010 (-0.366)	0.944** (2.147)
ROA	-0.182 (-1.363)	-3.124 (-1.589)
GROWTH	0.176*** (2.631)	3.260*** (3.096)
DUAL	0.028*** (3.258)	0.269** (2.133)
TOBINSQ	-0.004 (-0.968)	-0.064 (-1.393)
AGE	-0.096** (-2.230)	0.219 (0.326)
TOP1	0.107*** (2.621)	1.228* (1.776)
SOE	0.020 (1.134)	0.368 (1.318)
GDP	0.113*** (2.684)	-0.027 (-0.040)
Year FE	YES	YES
Firm FE	YES	YES
Observations	23,136	23,136
Adjusted R ²	0.343	0.007
Kleibergen- Paap rk LM	864.312	
Cragg- Donald Wald F statistic	1467.637	

4.3.4. High-dimension fixed-effect regression

Based on baseline model (Eq. (1)), we further include firm, year, industry and city fixed effects in regression.⁹ In Table 7, the coefficient of BR on GI is significantly positive, also suggesting baseline regression results are reliable.

⁹ This comprehensive approach captures and controls for various sources of heterogeneity, reducing the impact of omitted variable bias and other potential biases, and then enhances the accuracy of regression results.

Table 7

Results of high-dimension fixed-effect regression.

	(1)
	GI
BR	0.460*** (3.599)
SIZE	0.053 (1.058)
LEV	1.023 (1.524)
ROA	-3.404 (-1.320)
GROWTH	3.537*** (2.626)
DUAL	0.336* (1.959)
TOBINSQ	-0.073 (-1.215)
AGE	0.241 (0.206)
TOP1	1.592 (1.358)
SOE	0.495 (1.109)
GDP	0.392 (0.230)
Constant	-1.087 (-0.056)
Year FE	YES
Firm FE	YES
Industry FE	YES
City FE	YES
Observations	23,136
Adjusted R ²	0.602

4.3.5. Excluding samples during the COVID-19 periods

After excluding the sample during the global COVID-19 pandemic periods, we performed a regression and re-estimate the coefficients in the model, and the results confirm the baseline regression results is robust. The results are shown in Column (1)

of Table 8.

Table 8

Results of excluding samples during the COVID-19 periods.

	(1)
	GI
BR	0.656*** (4.503)
SIZE	0.034 (0.718)
LEV	0.437 (0.626)
ROA	-6.048** (-2.067)
GROWTH	4.551*** (3.103)
DUAL	0.564*** (2.843)
TOBINSQ	-0.165** (-2.431)
AGE	0.506 (0.441)
TOP1	0.850 (0.695)
SOE	0.482 (1.033)
GDP	0.472 (0.472)
Constant	-2.676 (-0.240)
Year FE	YES
Firm FE	YES
Observations	16,347
Adjusted R ²	0.019

5. Further test

5.1 Moderating effect analysis

5.1.1 Managerial shareholdings

Results in columns (1) of Table 9 show that BR positively influences GI. And the positive coefficient of the interaction term (BR*MS) indicates that the positive impact of biodiversity risk on firms' green investment is strengthened as managerial shareholdings increase. This finding **supports Hypothesis 2** and suggests that managerial shareholdings provide powerful incentives for undertaking strategic green investments in response to biodiversity risk.

5.1.2 Corporate ESG performance

As shown in column (2) of Table 9, BR exhibits a positive association with GI. And the coefficient of the interaction term (BR*ESG) is positive, indicating that the positive impact of biodiversity risk on firms' green investment is strengthened as corporate ESG performance improves. This result provides empirical **support for Hypothesis 3**, and suggests that high ESG ratings can enable firms cope with external risk such as ecological risk more effectively.

Table 9

Results of moderating test.

	(1)	(2)
	GI	GI
BR	0.497*** (3.671)	0.353*** (3.105)
MS	0.015 (0.815)	
BR*MS	0.018* (1.918)	
ESG		0.016 (0.271)
BR*ESG		0.090*** (3.465)
Constant	-0.006 (-0.000)	0.903 (0.080)
Controls	YES	YES
N	17,760	23,203
Year FE	YES	YES
Firm FE	YES	YES
Adjusted R ²	0.018	0.017

5.2 Heterogeneity analysis

5.2.1 Property-rights heterogeneity

Based on property rights, state-owned enterprises (SOEs) and non-state-owned enterprises (non-SOEs) differ in their business goals and the regulatory landscapes they navigate. Therefore, the impact of biodiversity risk on the green investment of these two types of enterprises may differ.

As shown in Table 10, columns (1) and (2), the coefficients of biodiversity risk (BR) are 0.637 and 0.383 respectively, and both are statistically significant. These results suggest that SOEs exhibit a stronger response to biodiversity risk in terms of green investment compared to non-SOEs. This disparity can be attributed to several factors. First, SOEs generally operate under more rigorous policy oversight and face greater public scrutiny, which motivates them to proactively engage in environmentally responsible practices. Second, their relatively superior decision-making processes and strong institutional support enable faster adaptation to environmental regulations and more readily obtain incentives such as tax benefits and green financing. Third, SOEs often possess greater access to resources and technological capabilities, allowing them to implement more advanced biodiversity conservation strategies. Taken together, these characteristics contribute to SOEs' greater responsiveness to biodiversity risks and their stronger commitment to green investment.

Therefore, SOEs have played a more active role in biodiversity conservation initiatives, resulting in significantly higher levels of green investment compared to non-SOEs.

5.2.2 Pollution-intensive heterogeneity

Firms in different industries may exhibit heterogeneous responses to biodiversity risk due to differences in environmental exposure, regulatory pressure, and operational characteristics. To examine whether the effect of biodiversity risk on green investment varies between pollution-intensive industries and non-pollution-intensive industries,¹⁰ we conduct a heterogeneity analysis by dividing the sample accordingly.

As shown in Table 10, columns (3) and (4), the coefficients of biodiversity risk (BR) are 0.539 (significant at the 10% level) and 0.441 (significant at the 1% level), respectively. The results indicate a potentially stronger economic impact of biodiversity risk on green investment in pollution-intensive industries. We identify three primary explanations for this finding. Firstly, pollution-intensive industries—such as

¹⁰ The classification of firms into pollution-intensive and non-pollution-intensive industries follows the *Guidelines for Industry Classification of Listed Companies* revised by the China Securities Regulatory Commission (CSRC) in 2012 and the *List of Industry Classification Management for Environmental Audit of Listed Companies* issued by the Ministry of Environmental Protection in 2008. Due to space constraints, the detailed classification and corresponding industries are provided in Appendix Table A.3.

manufacturing, mining, and chemicals sectors—typically have high environmental footprints and are often located in ecologically sensitive areas. Therefore, they are more directly exposed to biodiversity loss, which may drive greater investment in mitigation and restoration efforts. Second, these firms face stronger regulatory oversight and public scrutiny due to their potential negative impacts on ecosystems, prompting them to proactively increase green investment to comply with regulations, avoid penalties, and maintain social legitimacy. Third, biodiversity risk can pose significant operational challenges—such as resource scarcity and supply chain disruptions—to pollution-intensive firms, thereby providing stronger incentive to invest in green technologies and sustainable practices to ensure long-term business continuity.

This highlights the greater responsiveness of pollution-intensive industries to biodiversity risk, as they respond with greater green investment compared to their less-polluting counterparts.

Table 10

Results of heterogeneity analysis

	Property-rights heterogeneity		Pollution-intensive heterogeneity	
	(1)	(2)	(3)	(4)
	SOEs	non-SOEs	Pollution-intensive industries	Non-pollution-intensive industries
	GI	GI	GI	GI
BR	0.637*** (3.224)	0.383** (2.304)	0.539* (1.950)	0.441*** (3.158)
Constant	-15.389 (-1.064)	28.446 (1.426)	-9.095 (-0.331)	7.449 (0.661)
Controls	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Observations	9,804	13,399	5,473	17,730
Adjusted R ²	0.022	0.016	0.021	0.016

6. Conclusion

This study provides robust evidence that biodiversity risk promotes firms' green investment, supporting the strategic growth option theory. Furthermore, moderating effect analysis indicates that managerial shareholdings and corporate ESG performance can enhance the positive effect of biodiversity risk on firms' green investment. Additionally, the positive impact of biodiversity risk on green investment is more pronounced in SOEs and pollution-intensive industries.

Based on our findings, we offer practical implications for policymakers and corporate managers.

For policymakers: First, implement differentiated and targeted green incentives. Our finding that state-owned enterprises (SOEs) and firms in pollution-intensive industries are more responsive to biodiversity risk suggests that uniform policy instruments may underperform. Instead, policymakers are encouraged to design and implement differentiated incentive mechanisms. For example, offering preferential green credit lines, higher subsidies for green technology adoption, or targeted environmental tax relief specifically for these two groups could maximize policy effectiveness and accelerate the green transition where it is most impactful and urgently needed, thereby fostering a more resilient and competitive green economy. Second, strengthen governance and disclosure mandates. Given that managerial shareholdings and ESG performance significantly amplify green investment responses, regulators should promote governance structures that align managerial incentives with long-term sustainability. This includes mandating comprehensive, standardized biodiversity-related disclosures aligned with Taskforce on Nature-related Financial Disclosures (TNFD). Additionally, financial authorities could encourage or require that executive compensation be partially tied to verifiable environmental performance metrics, thereby institutionalizing internal mechanisms that foster proactive risk response.

For corporate managers: First, reframe biodiversity risk as a strategic growth opportunity. Managers should move beyond viewing biodiversity risk solely as a regulatory burden. Our findings lend strong support to the strategic growth option theory, implying that managers should adopt the strategic lens of this theory and treat

risk as a potential market opportunity. Following the logic of this framework, proactive green investment is a crucial tool for securing first-mover advantages, enhancing brand reputation, and building a sustainable competitive edge in an increasingly eco-conscious market. Second, deeply integrate sustainability goals into corporate governance and core strategy. This requires a two-pronged approach. On one hand, Boards and Compensation Committees should consider aligning executive incentive structures more closely with long-term green innovation and sustainability targets. Our results indicate that when managers have a direct financial stake in the firms' long-term success, their firms are better equipped to respond proactively and decisively to emerging ecological challenges like biodiversity risk, helping to overcome the managerial short-termism that can often stifle such crucial, forward-looking investments. On the other hand, integrating green investment with ESG strategy is important. Managers should proactively invest in enhancing corporate ESG performance, particularly on the Environmental (E) pillar. This involves more than just reporting; it means embedding robust environmental monitoring and management systems into core operations. As our findings show, a strong ESG track record does not just build stakeholder trust—it strengthens the firms' capacity to convert external risks into strategic green investments. This creates a virtuous cycle where better data improves ESG scores, which in turn can lower the cost of capital and unlock further green investment opportunities.

Future research could explore the following areas:

First, future work could contribute by examining how different types of biodiversity risk, such as physical risks from resource scarcity versus transition risks from regulatory changes, uniquely influence corporate strategy like green investment. **Second**, researchers are encouraged to investigate other potential mediating or moderating factors, such as stakeholder pressure or executive environmental awareness. **Third**, future research could explore the heterogeneous impacts of biodiversity risk by disaggregating green investment into distinct categories (e.g., proactive 'technology-driven' initiatives versus reactive 'eco-restoration' projects), which would offer a more nuanced understanding of corporate greening strategies.

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Appendix

Table A.1

Biodiversity risk-related terms.

Biodiversity, ecosystem(s), ecology/ecological, habitat(s), species, (rain)forest(s), deforestation, fauna, flora, marine, tropical, freshwater, wetland, wildlife, coral, aquatic, desertification, carbon sink(s), ecosphere, and biosphere (He et al., 2024)

Table A.2**Variable definition**

Variables	Symbol	Definition	Source
Dependent variable	GI	Ln (1+firm's total capital expenditure related to environmental protection)	Manually collect
Independent variable	BR	A dummy variable developed by He et al. (2024), equals 1 if the frequency of biodiversity-related terms in a firm's annual report is more than two, and 0 otherwise.	He et al. (2024)
Control variables	SIZE	Total assets (in 10 billion)	CSMAR
	LEV	Total liabilities / total assets	CSMAR
	ROA	Return on assets	CSMAR
	GROWTH	$(ROE * Retention Rate) / (1 - ROE * Retention Rate)$	CSMAR
	DUAL	A dummy variable, equals 1 if the chairman and general manager positions are held by the same person, and 0 otherwise	CSMAR
	TOBINSQ	Total market value / total assets value	CSMAR
	AGE	Natural logarithm of the firm's age in years since its establishment	CSMAR
	TOP1	Shareholding ratio of the largest shareholders	CSMAR
	SOE	A dummy variable, equals 1 if the enterprise is state-owned, and 0 otherwise.	CSMAR
	GDP	Logarithm of provincial GDP per capita	CSMAR
Moderating variables	MS	Ln (1+ Total number of shares held by members of firm's directors, senior management, and supervisors)	CSMAR
	ESG	Corporate ESG performance	Sino-Securities Index (SNSI) ESG score

Table A.3**Category of pollution-intensive industries and non-pollution-intensive industries**

Category	Detail information
Pollution-intensive industries	Coal mining and washing (B06), Oil and natural gas mining (B07), Ferrous metal mining (B08), Non-ferrous metal mining (B09), Textiles (C17), Leather, fur, feathers and their products and footwear (C19), Paper and paper products (C22), Petroleum processing, coking and nuclear fuel processing (C25), Manufacture of chemical raw materials and chemical products (C26), Manufacture of chemical fibers (C28), Manufacture of rubber and plastic products (C29), Manufacture of non-metallic mineral products (C30), Ferrous metal smelting and rolling processing (C31), Non-ferrous metal smelting and rolling processing (C32), and Electricity and heat production and supply (D44).
Non-pollution-intensive industries	All other industry categories are considered non-pollution-intensive.

Author contribution statement

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I confirm that I am the sole author and corresponding author of this manuscript, and I will serve as the single point of contact throughout the editorial process.

Author Contribution:

GangWang: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing-Original Draft, Writing-Review & Editing.